General Feedback for Graded Formative Sheet 1

13 October 2026

There were some issues where variables appeared without being introduced. If you use an expression gxg^{-1}, say, then you should begin by stating what each of the variables refer to and (if relevant) how they are quantified (for all / there exists). For instance, in *proving* A is normal in G, you may start by writing, "Let $g \in G$ and let $a \in A$" and end after you have proved, "$gag^{-1} \in A$". If you are *using* a condition that A is normal in G, then you would write "For every $g \in G$ and for every $a \in A$, we have $gag^{-1} \in A$".

- [Proving that $H \cap K$ is a subgroup of G] Ensure you verify *all* three properties of a subgroup. Some submissions didn't show that $H \cap K$ is non empty. Other submissions had missing steps. For example, it isn't sufficient to *state* that $e_G \in H \cap K$, you must give a reason. For example, you could state "as H and K are subgroups of G, we have $e_G \in H$ and $e_G \in K$. So $e_G \in H \cap K$.
- [Normal subgroups] Several answers either gave an incorrect definition of normal subgroup or *used* a correct definition *incorrectly*. For example, it isn't (in general) true that if N is a normal subgroup of G and if $x \in G$ then $xn = nx$ for every $x \in G$ and every $n \in N$. For this question, the simplest criterion to use is that for N to be normal in G means that for every $x \in G$ and for every $n \in N$, one has $xnx^{-1} \in N$.